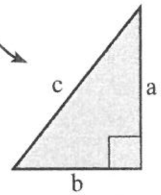
Pythagoras' Theorem

In any right-angled triangle:
The square on the hypotenuse is equal to the sum of the squares on the other two sides.

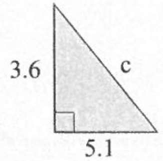
The hypotenuse is the longest side.
It is opposite the right-angle (90°).



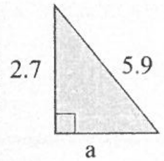
$c^2 = a^2 + b^2$

$a^2 + b^2 = c^2$

Exercise 17.1
Find the length of the unknown in each of the following (round to two decimal places):

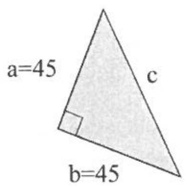


$c^2 = a^2 + b^2$
$c^2 = 3.6^2 + 5.1^2$
$c^2 = 38.97$
$c = \sqrt{38.97}$
$\underline{c = 6.24}$



$a^2 + b^2 = c^2$
$a^2 + 2.7^2 = 5.9^2$
$a^2 = 5.9^2 - 2.7^2$
$a^2 = 27.52$
$a = \sqrt{3.87}$
$\underline{a = 5.25}$

1



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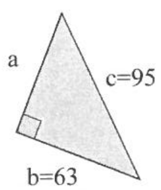

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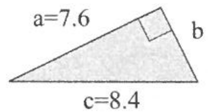
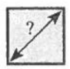


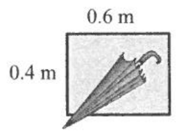
Rounding to two decimal places, first look at the third decimal place:

56.231694 ↑ less than 5 thus **56.23**	27.01769 ↑ 5 or more thus **27.02**	1.07276 ↑ less than 5 thus **1.07**	4.79634216 ↑ 5 or more thus **4.80**

4 What is the length of the diagonal of a 5 m square?



5 Will a 0.8 m umbrella fit in a 0.6 m by 0.4 m case?



6 A 2.8 m ladder is to be laid against a wall so that the top of the ladder is 2 m up the wall. How far out from the base of the wall should the ladder be placed?

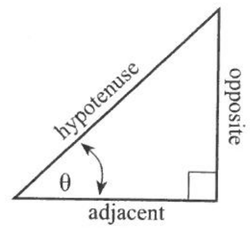
7 A 45 m tower is supported by guy wires. The guy wires are attached to the top of the tower and anchored to the ground 18 m out from the base of the tower. How long are the guy wires?

The Tan Ratio

Trigonometry was developed thousands of years ago to solve the many problems in surveying, engineering, architecture, astronomy, etc, etc, etc.

Trigonometry n. branch of mathematics dealing with the relationships between angles and sides of triangles.

$\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}}$



Opposite is 'opposite' the angle.

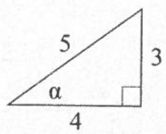
Adjacent means 'next to' the angle.

Greek letters:
α is alpha
θ is theta

Exercise 17.2
Find tan α and the size of the angle α (round to two decimal places):

3 is opposite α

4 is adjacent to α



$\tan\alpha = \dfrac{\text{opposite}}{\text{adjacent}}$

$\tan\alpha = \dfrac{3}{4} = \underline{0.75}$

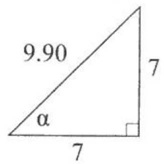
$\alpha = \tan^{-1} 0.75$
$\underline{\alpha = 36.87°}$

Use your calculator:
[2ndF] [$\tan^{-1}$] [0.75] [=]

$\tan^{-1}$ means 'an angle whose tan is'.
Thus $\tan^{-1}$ 0.75 means 'an angle whose tan is 0.75' (which is 36.87°).

Make sure your calculator is on **degrees**.

1



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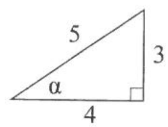

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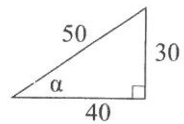

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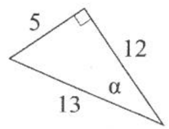

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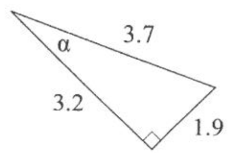

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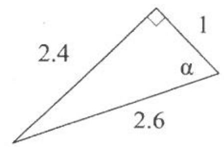
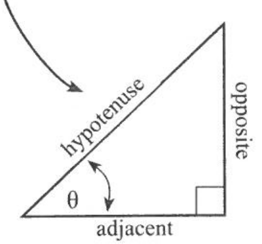


The Sine Ratio

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

The hypotenuse is the longest side.
It is opposite the right-angle (90°).

Opposite is 'opposite' the angle.

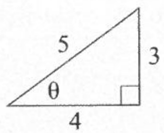


Exercise 17.3

Find sin θ and the size of the angle θ (round to two decimal places):

3 is opposite θ

5 is the hypotenuse



$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\sin\theta = \frac{3}{5} = 0.6$$

$$\theta = \sin^{-1} 0.6$$

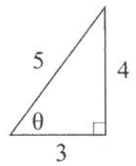
$$\underline{\theta = 36.87°}$$

Use your calculator: 2ndF sin⁻¹ 0.6 =

sin⁻¹ means 'an angle whose sine is'.
Thus $\sin^{-1} 0.6$ means 'an angle whose sine is 0.6' (which is 36.87°).

Make sure your calculator is on **degrees**.

1



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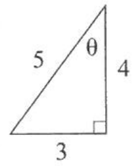

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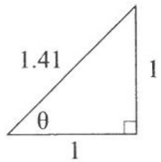

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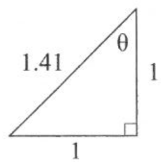

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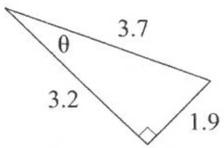

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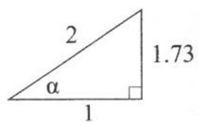
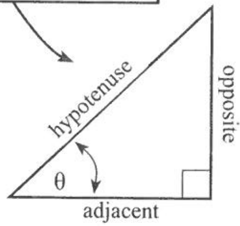


The Cos Ratio

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

The hypotenuse is the longest side.
It is opposite the right-angle (90°).



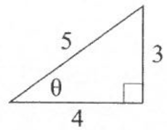
Adjacent means 'next to' the angle.

Exercise 17.4

Find cos θ and the size of the angle θ (round to two decimal places):

4 is adjacent θ

5 is the hypotenuse



$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\cos\theta = \frac{4}{5} = 0.8$$

$$\theta = \cos^{-1} 0.8$$

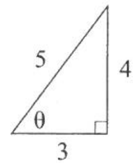
$$\underline{\theta = 36.87°}$$

Use your calculator: 2ndF cos⁻¹ 0.8 =

cos⁻¹ means 'an angle whose cosine is'.
Thus $\cos^{-1} 0.8$ means 'an angle whose cos is 0.8' (which is 36.87°).

Make sure your calculator is on **degrees**.

1



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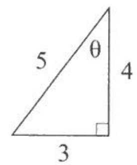

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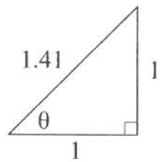

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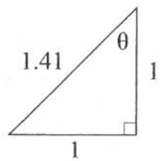

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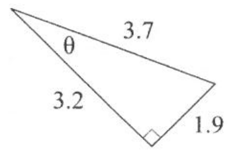

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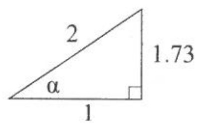


Trigonometry

Trigonometry can be used to find a side after knowing an angle and a side in a right-angled triangle.

The trick is which one?

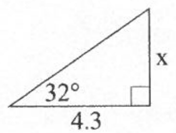
$\sin\alpha = \frac{\text{opposite}}{\text{hypotenuse}}$

$\cos\alpha = \frac{\text{adjacent}}{\text{hypotenuse}}$

$\tan\alpha = \frac{\text{opposite}}{\text{adjacent}}$

Exercise 17.5

Find x in each of the following right-angled triangles:



α = 32
adjacent = 4.3
opposite = x

$\sin\alpha\checkmark = \frac{\text{opposite}?}{\text{hypotenuse}}$

$\cos\alpha\checkmark = \frac{\text{adjacent}\checkmark}{\text{hypotenuse}}$

$\tan\alpha\checkmark = \frac{\text{opposite}?}{\text{adjacent}\checkmark}$ ← Use this one

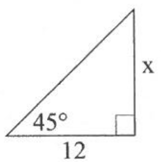
1 Write the knowns and unknown.
2 ✓ the knowns and ? the unknown.
3 The one to use is fully marked.

$\tan 32 = \frac{x}{4.3}$

$\tan 32 \times 4.3 = x$

$\underline{2.69 = x}$

1



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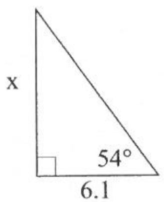

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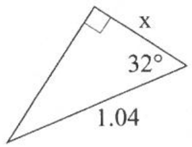

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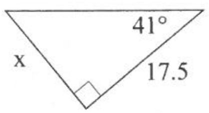

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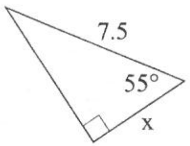
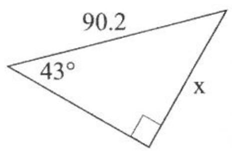
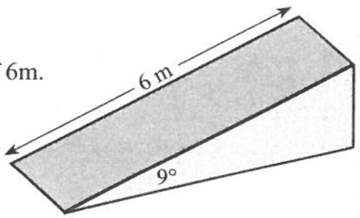


6



7 A ramp has an incline of 9° and a length of 6m. What is the length of the horizontal base?



Trigonometry is used millions and millions of times every day.

The trick is which one?

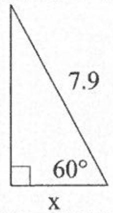
$\sin\alpha = \frac{\text{opposite}}{\text{hypotenuse}}$

$\cos\alpha = \frac{\text{adjacent}}{\text{hypotenuse}}$

$\tan\alpha = \frac{\text{opposite}}{\text{adjacent}}$

Exercise 17.6

Find x in each of the following right-angled triangles:



α = 60
hypotenuse = 7.9
adjacent = x

$\sin\alpha\checkmark = \frac{\text{opposite}}{\text{hypotenuse}\checkmark}$

$\cos\alpha\checkmark = \frac{\text{adjacent}?}{\text{hypotenuse}\checkmark}$ ← Use this one

$\tan\alpha\checkmark = \frac{\text{opposite}}{\text{adjacent}?}$

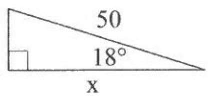
1 Write the knowns and unknown.
2 ✓ the knowns and ? the unknown.
3 The one to use is fully marked.

$\cos 60 = \frac{x}{7.9}$

$\cos 60 \times 7.9 = x$

$\underline{3.95 = x}$

1



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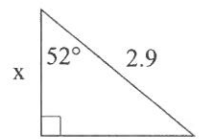

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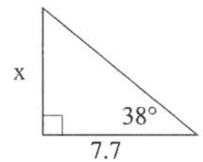

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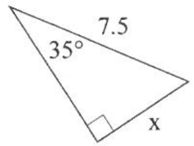

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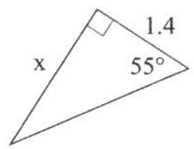

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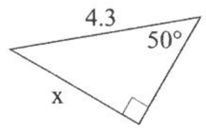


7 The beam of a see-saw is 3.6 m long.
When one end is resting on the ground,
the angle the beam makes with the ground is 17°.
How far above the ground is the other end?

Trigonometry

We have made use of the ratios of corresponding sides of right-angled triangles for thousands of years.

The trick is which one?

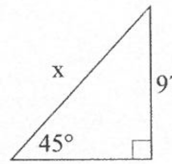
$\sin\alpha = \frac{\text{opposite}}{\text{hypotenuse}}$

$\cos\alpha = \frac{\text{adjacent}}{\text{hypotenuse}}$

$\tan\alpha = \frac{\text{opposite}}{\text{adjacent}}$

Exercise 17.7

Find x in each of the following right-angled triangles:



$\alpha = 45$
opposite = 97
hypotenuse = x

$\sin\alpha\checkmark = \frac{\text{opp?site}}{\text{hypotenuse}\checkmark}$ ← Use this one

$\cos\alpha\checkmark = \frac{\text{adjacent}}{\text{hypotenuse}\checkmark}$

$\tan\alpha\checkmark = \frac{\text{opp?site}}{\text{adjacent}}$

$\sin 45 = \frac{97}{x}$

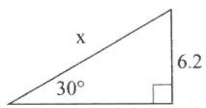
$\sin45 \times x = 97$

$x = \frac{97}{\sin 45}$

$\underline{137.18 = x}$

1 Write the knowns and unknown.
2 ✓ the knowns and ? the unknown.
3 The one to use is fully marked.

1



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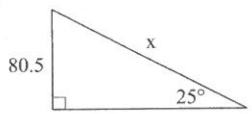

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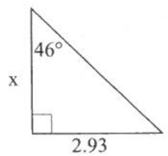

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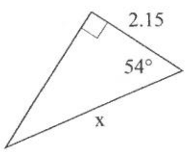

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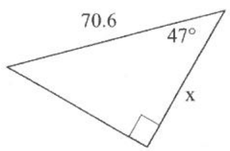

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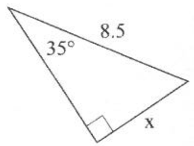
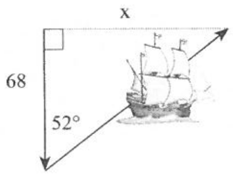


7 A sailing boat sails south for 68 km, then on a bearing of 52°T until it is due east of its starting point.
How far is the boat from its starting point?



Trigonometry can be used to find an **angle** after knowing two sides in a right-angled triangle.

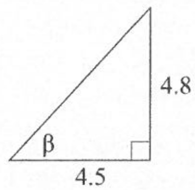
Greek letters:
α is alpha
β is beta
θ is theta

Pythagoras' Theorem can be used to find the third side after knowing two sides in a right-angled triangle.

The angles in a triangle sum to 180°

Exercise 17.8

Find β in each of the following right-angled triangles:



opposite = 4.8
adjacent = 4.5
angle = β

$\sin?\alpha = \frac{\text{opp}\checkmark\text{site}}{\text{hypotenuse}}$

$\cos?\alpha = \frac{\text{adj}\checkmark\text{cent}}{\text{hypotenuse}}$

$\tan?\alpha = \frac{\text{opp}\checkmark\text{site}}{\text{adj}\checkmark\text{cent}}$ ← Use this one

1 Write the knowns and unknown.
2 ✓ the knowns and ? the unknown.
3 The one to use is fully marked.

$\tan\beta = \frac{4.8}{4.5}$

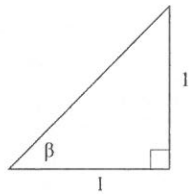
$\tan\beta = 1.07$

$\beta = \tan\text{-}1(1.07)$

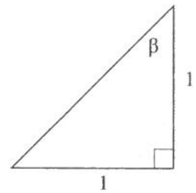
$\underline{\beta = 46.94°}$

Use your calculator:
$\boxed{\text{2ndF}}\ \boxed{\tan^{-1}}\ \boxed{1.07}\ \boxed{=}$

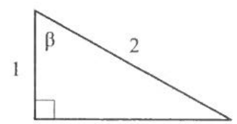
1



2



3



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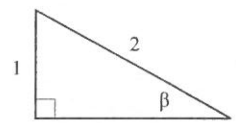
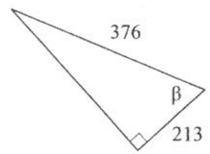
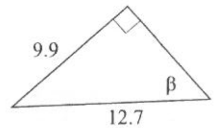


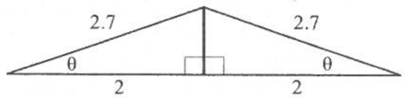
5



6



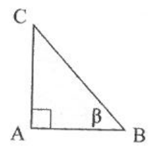
7 What is the pitch, θ, of the roof?



Chapter Review 1

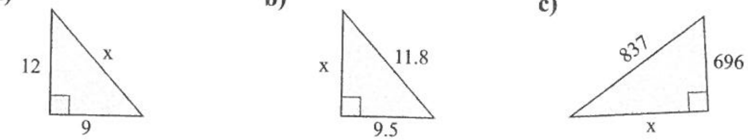
Exercise 17.14

1 For the adjacent triangle, name:
- **a)** the hypotenuse.
- **b)** the side adjacent to the angle.
- **c)** the side opposite the angle.



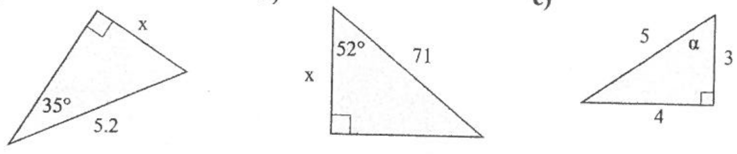
2 Use Pythagoras' Theorem to find the unknown:

a) **b)** **c)**



3 Find the unknown in each of the following triangles:

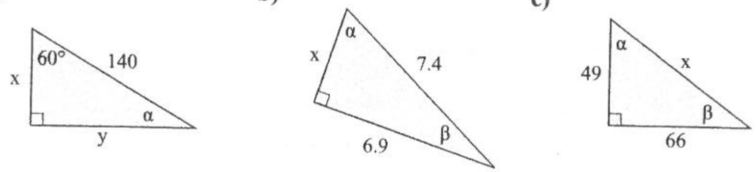
a) **b)** **c)**



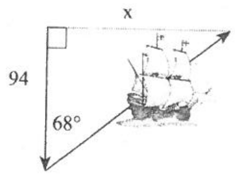
Solve means 'find **all** unknowns'. (3 sides, 3 angles)

4 Solve the following triangles:

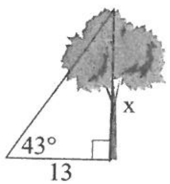
a) **b)** **c)**



5 A sailing boat sails south for 94 km, then on a bearing of 68°T until it is due east of its starting point.
How far is the boat from its starting point?
How far has the boat travelled?



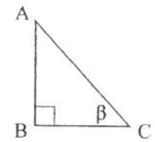
6 13 m out from the base of a tree, a clinometer measures the angle of elevation to the top of the tree as 43°. Find the height of the tree.



Chapter Review 2

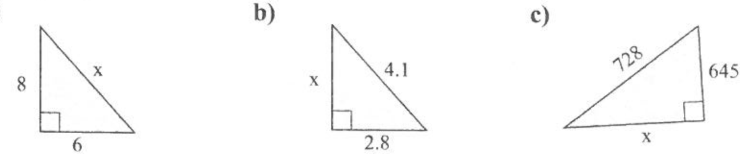
Exercise 17.15

1 For the adjacent triangle, name:
- **a)** the hypotenuse.
- **b)** the side adjacent to the angle.
- **c)** the side opposite the angle.



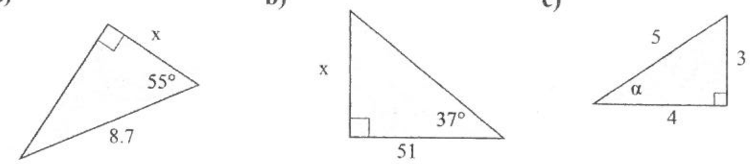
2 Use Pythagoras' Theorem to find the unknown:

a) **b)** **c)**



3 Find the unknown in each of the following triangles:

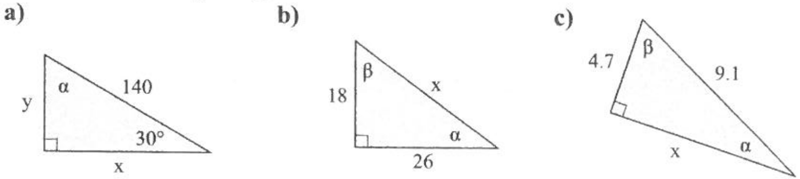
a) **b)** **c)**



Solve means 'find **all** unknowns'. (3 sides, 3 angles)

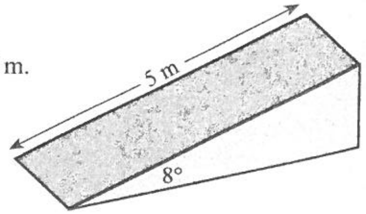
4 Solve the following triangles:

a) **b)** **c)**



5 The beam of a see-saw is 3.2 m long.
When one end is resting on the ground,
the angle the beam makes with the ground is 23°.
How far above the ground is the other end?

6 A ramp has an incline of 8° and a length of 5 m.
What is the length of the horizontal base?



7c $3\sqrt{2}$ or 4.24 **8a** Midpoint of AC and BD is same (3.5,3) **8b** m of AB and CD is same 0.43 thus parallel **8c** AB=CD=7.62 and AD=BC=5.39

Exercise 16.10 **2** $m=(y_2-y_1)/(x_2-x_1)$ **3** $d=\sqrt{((x_2-x_1)^2+(y_2-y_1)^2)}$ **4** mid=$((x_1+x_2)/2,(y_1+y_2)/2)$ **5** m=1 **6** d=$\sqrt{8}$ or $2\sqrt{2}$ **7** mid=(2,3) **8** 1/6 **9** $A=\pi r^2$ **10** 1

Exercise 16.11 **2** $m=(y_2-y_1)/(x_2-x_1)$ **3** $d=\sqrt{((x_2-x_1)^2+(y_2-y_1)^2)}$ **4** mid=$((x_1+x_2)/2,(y_1+y_2)/2)$ **5** m=1 **6** d=$\sqrt{18}$ or $3\sqrt{2}$ **7** mid=(3.5,2.5) **8** 1/6 **9** $A=\pi r^2$ **10** 1

Exercise 16.12 **2** $m=(y_2-y_1)/(x_2-x_1)$ **3** $d=\sqrt{((x_2-x_1)^2+(y_2-y_1)^2)}$ **4** mid=$((x_1+x_2)/2,(y_1+y_2)/2)$ **5** m=0.6 **6** d=$\sqrt{34}$ **7** mid=(3.5,2.5) **8** 2/6=1/3 **9** $A=\pi r^2$ **10** 0.6

Exercise 16.13 **1** 30 **2** 3 **3** 0.3 **4** 0.03 **5** 400 **6** 4 **7** 0.04 **8** 0.0004 **9a** m=1 **9b** m=0.5 **9c** m=0.33 **10** 53°,90° **11a** 6units^2 **11b** 9units^2 **12** Q(7,7),P(8,4),mQP=$^-3$, mAB=$^-3$, thus parallel **13** $5\sqrt{3}$

Exercise 16.14 **1a** $\begin{array}{r} 345 \\ 345 \\ 345 \\ \underline{345} \\ \underline{1380} \end{array}$ **1b** $\begin{array}{r} 345 \\ 345 \\ 345 \\ \underline{345} \\ 1380 \\ \underline{345} \\ \underline{1725} \end{array}$ **1c** $\begin{array}{r} 9567 \\ \underline{1085} \\ \underline{10652} \end{array}$

Exercise 16.15 **1a** 7.21 **1b** 8.60 **1c** 5 **1d** 4.47 **1e** 2.24 **2a** (-1,1) **2b** (0.5,0.5) **2c** (3,2) **2d** (1,2) **2e** (-0.5,0.5) **3a** 1.5 **3b** -2.33 **3c** 1 **3d** -0.67 **3e** -0.57 **4a** m of AB and CD=1.33 **4b** AB=CD=5, AD=BC=6.71

Exercise 16.16 **1a** 8.06 **1b** 8.25 **1c** 2.83 **1d** 8.94 **1e** 5 **2a** (3,0.5) **2b** (-2.5,0.5) **2c** (3,5) **2d** (0,1) **2e** (-1,1.5) **3a** 1.4 **3b** -1.33 **3c** 1 **3d** -0.6 **3e** 1.14 **4a** m of AB and CD=0.2 **4b** AB=CD=5.10, AD=BC=3.61

Exercise 17.1 **1** $45\sqrt{2}$ or 63.64 **2** 71.11 **3** 3.58 **4** $5\sqrt{2}$ or 7.07 **5** no **6** 1.96m **7** 48.47m

Exercise 17.2 **1** tanα=1, α=45° **2** tanα=0.75, α=36.87° **3** tanα=0.75, α=36.87° **4** tanα=0.42, α=22.78° **5** tanα=0.59, α=30.54° **6** tanα=2.4, α=67.38°

Exercise 17.3 **1** sinα=0.8,α=53.13° **2** sinα=0.6,α=36.87° **3** sinα=0.71,α=45.23° **4** sinα=0.71,α=45.23° **5** sinα=0.51,α=30.66° **6** sinα=0.87,α=60.46°

Exercise 17.4 **1** cosα=0.6,α=53.13° **2** cosα=0.8,α=36.87° **3** cosα=0.71,α=44.77° **4** cosα=0.71,α=44.77° **5** cosα=0.86,α=30.68° **6** cosα=0.5,α=60°

Exercise 17.5 **1** 12 **2** 8.40 **3** 0.88 **4** 15.21 **5** 4.30 **6** 61.52 **7** 5.93m

Exercise 17.6 **1** 47.55 **2** 1.79 **3** 6.02 **4** 4.30 **5** 2.00 **6** 3.29 **7** 1.05m

Exercise 17.7 **1** 12.4 **2** 190.48 **3** 2.83 **4** 3.66 **5** 48.15 **6** 4.88 **7** 87.04km

Exercise 17.8 **1** 45° **2** 45° **3** 60° **4** 30° **5** 55.49° **6** 51.22° **7** 42.21°

Exercise 17.9 **2** 0.8 **3** 0.6 **4** 1.33 **5** 30°,90° **6** 1 **7** $\sqrt{8}$ or $2\sqrt{2}$ **8** (2,3) **9** 5.4×10^4 **10** $C=2\pi r$ or $C=\pi D$

Exercise 17.10 **2** 12/13 **3** 5/13 **4** 12/5 or 2.4 **5** 60°,90° **6** 2 **7** $\sqrt{5}$ **8** (2.5,3) **9** 6.2×10^5 **10** $A=\pi r^2$

Exercise 17.11 **2** $1/\sqrt{2}$ **3** $1/\sqrt{2}$ **4** 1 **5** 45°,90° **6** 1 **7** $\sqrt{8}$ or $2\sqrt{2}$ **8** (2,3) **9** 4.8×10^6 **10** $C=2\pi r$ or $C=\pi D$

Exercise 17.12 **1a** 60° **1b** 45° **1c** 30° **2a** sinα=$2/\sqrt{3}$,cosα=$1/\sqrt{3}$ **2b** sinα=$1/\sqrt{2}$,cosα=$1/\sqrt{2}$ **2c** sinα=$1/\sqrt{3}$,cosα=$2/\sqrt{3}$ **3a** 30° **3b** 60° **3c** 45° **4a** 26 **4b** 15

Exercise 17.13 **1** Three hundred and forty-two billion four hundred and eighty-two million eight hundred and seventy-three thousand seven hundred and ten **2** 0.0001 **3** 21,34 **4**

Exercise 17.14 **1a** BC **1b** AB **1c** AC **2a** 15 **2b** 7.00 **2c** 464.92 **3a** 2.98 **3b** 43.71 **3c** 53.13° **4a** x=70,y=121.24,α=30° **4b** x=2.67,α=68.82°,β=21.18° **4c** x=82.20,α=53.41°,β=36.59° **5** 232.66km **6** 12.12m

Exercise 17.15 **1a** AC **1b** BC **1c** AB **2a** 10 **2b** 2.99 **2c** 337.58 **3a** 4.99 **3b** 38.43 **3c** 36.87° **4a** x=121.24,y=70,α=60° **4b** x=31.62,α=34.70°,β=55.30° **4c** x=7.79,α=31.10°,β=58.90° **5** 1.25m **6** 4.95m